An update to the global critical habitat screening layer

Sebastian Dunnett[1], **Alfred Muge**[1], **Alex Ross**[1], **Joe Turner**[2], **Neil Burgess**[1], **and Matt Jones**[1]

[1]United Nations Environment Programme World Conservation Monitoring Centre, 219 Huntingdon Road, Cambridge
[2]The Biodiversity Consultancy, 3E King's Parade, Cambridge

ABSTRACT

The International Finance Corporation (IFC) defines Critical Habitat as areas of high biodiversity value in their Performance Standard 6 (PS6), requiring projects located in Critical Habitat to achieve a net gain of biodiversity. IFC's definition of Critical Habitat has since been frequently used by the finance sector and businesses to identify areas of high biodiversity value. Here we present an update to previous global screening layers of Critical Habitat in the marine and terrestrial realms, derived from global spatial datasets covering the distributions of 54 biodiversity features aligned with guidance provided by the IFC. As with previous layers, each biodiversity feature is categorised as 'Likely' or 'Potential' Critical Habitat based on: 1. Alignment between the biodiversity feature and the IFC Critical Habitat definition; and 2. Suitability of the spatial resolution for indicating a feature's presence on the ground. This analysis indicates that a total of 51.6 million km^2 (10%) and 16.6 million km^2 (3%) of the globe can be considered as Likely and Potential Critical Habitat respectively, while the remaining 441.8 million km^2 (87%) did not overlap with any of the biodiversity features assessed. This represents a significant increase to the 26.2 million km^2 (5%) and 10 million km^2 (2%) Likely and Potential Critical Habitat identified previously. Classification of Likely Critical Habitat was dominated by Important Bird and Biodiversity Areas, Intact Forest Landscapes, and protected areas; Potential Critical Habitat by Important Marine Mammal Areas (IMMAs) and ranges of species classified as Vulnerable by the International Union for Conservation of Nature (IUCN) Red List of Threatened Species. Our results can aid businesses to prioritise impact avoidance by screening potential development sites at an early project stage for biodiversity features.

Background & Summary

More than US$60 trillion of infrastructure spending is likely required to meet 2040 societal goals, with an additional 1.2 million km^2 urbanised land by 2030, an additional 3-4.7 million km of roads by 2050[1,2,3]. During the relatively modest infrastucture expansion of the late twentieth and early twenty-first centuries, biodiversity has continued to decline precipitously, with governments collectively failing every single one of the Aichi Biodiversity Targets set by the United Nations (UN) Convention on Biological Diversity (CBD) in 2011. Much of the potential expansion of infrastructure is predicted to occur in some of the world's highest integrity ecosystems[4,5,6,7].

Business and biodiversity para[8]: suggests more accountable businesses will direct corporate action with Critical Habitat screening. Citigroup uses[9]: half of marine Critical Habitat not formally protected.

In 2012, the International Finance Corporation (IFC; a member of the World Bank Group) developed *Performance Standard 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources* (hereafter IFC PS6), one of eight Performance Standards any client of the organisation must meet throughout the life of an investment[10]. PS6 draws heavily on robust conservation datasets such as the World Database on Protected Areas (WDPA) and the IUCN Red List of Threatened Species (hereafter IUCN Red List)[11,12,13]. The standard is seen as the benchmark for sustainable investment with 130 financial institutions following its social and environmental classification process as signatories to the Equator Principles[14]. Critical Habitat is defined by IFC using the following criteria:

1. Critically Endangered or Endangered species
2. Endemic and/or restricted-range species
3. Globally significant concentration of migratory or congregatory species
4. Highly threatened and/or unique ecosystems
5. Key evolutionary processes

Previous work by **(author?)**[15] and **(author?)**[16] produced global 1 km^2 screening layers for the marine and terrestrial realms respectively. These studies identified global biodiversity feature data with sufficient resolution and alignment with the IFC PS6 criteria to produce global maps of Likely and Potential Critical Habitat. **(author?)** [15] classified 1.6% of the ocean as Likely Critical Habitat and 2.1% as Potential Critical Habitat. **(author?)** [16] identified 10% of the terrestrial surface as Likely Critical Habitat and 5% as Potential Critical Habitat. The screening layers, as well as the amalgamated marine plus terrestrial Critical Habitat screening map made available through the UN Environment Programme World Conservation Monitoring Centre's data portal, have been used in a number of subsequent studies. **(author?)** [17] analysed the impact of PS6 on biodiversity offset implementation. Many of the studies focus on the threat to biodiversity of China's Belt and Road Initiative, a programme to link sixty-five countries with a network of transport and energy infrastructure. Researchers have used the global Critical Habitat screening layer to estimate that 50% of loans intersect with Critical Habitat[18]. Full 49 Chinese-funded dams have a total of 149 km^2 Critical Habitat in close proximity (i.e. within 1 km) and dams funded by multilateral development banks (MDBs) have much lower average areas of Critical Habitat per area of dam at risk relative to Chinese-funded dams[19]. Researchers have also identified 12,000 km^2 of Critical Habitat within 1 km of the Belt and Road Initiative's linear infrastructure[20]. In a decarbonising world, 20% of mining locations in Africa trigger a Critical Habitat classification for something other than Great Ape habitat[21].

Between November 2018 and June 2019, IFC updated the 2012 PS6 Guidance Note that had accompanied the initial release[22]. The update significantly changed the Critical Habitat criteria to better align the standard with the then newly developed *Global Standard for the Identification of Key Biodiversity Areas* (KBAs) , Table 1. This meant that the original Critical Habitat global screening layers were now out of sync with the latest criteria. At the same time, many of the datasets that underpinned the original analyses have been updated multiple times since (e.g. the WDPA is updated monthly). Finally, many eligible datasets have been produced in the years since that meet the completeness and alignment criteria of the layer.

Table 1. Changes to the IFC PS6 criteria post-2018. Adapted from **(author?)** [23].

Criteria	GN 2012	GN 2018	Implications
Critically Endangered or Endangered species	Tier 1: 10% Tier 2: >0%	0.5%	Small increase in threshold Reduction in overall Potential Critical Habitat
Endemic and/or restricted-range species	Tier 1: 95% Tier 2: 1%	10%	Threshold significantly higher Reduction in overall Potential Critical Habitat
Globally significant concentrations of migratory or congregatory species	Tier 1: 95% Tier 2: 1%	1%	Threshold unchanged No change to overall Potential Critical Habitat
Highly threatened and/or unique ecosystems	Qualitative No thresholds	5%	New threshold
Key evolutionary processes	Qualitative No threshold	Qualitative No threshold	Unchanged

It is critical, for businesses to manage their biodiversity impacts in areas of high biodiversity value, to provide an updated Critical Habitat layer that fully aligns with the 2018/2019 update of PS6. In this Data Descriptor, we describe a new, updated global Critical Habitat screening layer. The new data incorporates updated datasets, as well as new sources identified through a comprehensive search for any that could meet the eligibility criteria. Most importantly, we describe and make available a methodology whereby the layer can be updated as and when new data are found or updated.

Methods

Critical Habitat criteria

IFC define Critical Habitat according to five criteria (Table 1) with associated thresholds for Criteria 1-4. The 2018/2019 update to the accompanying Guidance Note made a number of changes to the classification of Critical

Habitat of relevance to any updated methodology:

- **Changes to thresholds**: thresholds have been updated to better align with the Global KBA Standard[24]. Criteria 1, 3 and 4 now directly align with the standard, and Criteria 1-3 have streamlined thresholds that largely sit between the previous Tier 1 and Tier 2 thresholds. Quantitative thresholds have been added for Criterion 4, based on the developing IUCN Red List of Ecosystems[25].
- **Specified exceptional circumstances**: IFC now specify three circumstances that trigger special considerations.
  - Great Apes: the presence of Great Ape species is likely to trigger Critical Habitat regardless of thresholds. Where present, clients are expected to inform both IFC and the IUCN Species Survival Commission's Primate Specialist Group.
  - Unapprovable sites:
    * World Heritage Sites (Natural or Mixed).
    * Alliance for Zero Extinction sites.
- **Addition of IUCN Vulnerable species**: IFC added a threshold for Criteria 1 that now explicitly includes species classified as Vulnerable (VU) by the IUCN Red List where:
  - Globally important populations are present.
  - The loss of the population would lead to the species being upgraded to Endangered (EN) or Critically Endangered (CR).
  - The population in question would then trigger Criterion 1a (the area supports >=0.5% of the global population and >=5 reproductive units).

Data screening and classification

Using the updated criteria, we proceeded with data screening and classification as with previous studies[15,16]. Datasets were identified through expert knowledge and consultation, using the criteria adapted from **(author?)** [15]:

1. Direct relevance to one or more Critical Habitat criteria.
2. Global in extent.
3. Assembled using a standardised protocol.
4. The best available data for the biodiversity feature of interest.
5. Sufficiently high resolution to indicate presence of biodiversity on the ground at scales relevant to business operations.

Datasets that met our criteria were classified as one of three classes: Likely, Potential, and Unclassified. The classification was done by expert judgement, weighing the alignment with PS6 criteria and the likely presence on the ground ({#fig-class-grid}).

Data processing and spatial analysis

Spatial data analysis was conducted in **R** using predominatly the `terra` and `sf` packages. See Supplementary Information for more detailed methodology.

For the greatest accuracy, vector data were processed using the `sf` package and `S2` library. This allows spatial processing, e.g. intersections and buffers, using Great Circle distances. Data were prepared for the analysis in three steps: first, data were made valid on the sphere. Second, data were unioned to remove duplicates from point data and any overlaping area in polygon data. Finally, data were filtered as detailed by (a) their respective guidance, and (b) to produce the data subset required for the analysis. For example, Key Biodiversity Areas (KBAs) must first be filtered by those whose status is confirmed (`KbaStatus=="confirmed"`) and then by the criterion/criteria required to create the subset (e.g. sites designated under KBA Criterion B1). Not all input data were available in vector format. For raster data, preprocessing varies by source but generally involves aggregating to the correct resolution (0.008333 degrees) before reprojecting the data to a template raster in WGS 84.

Data are then converted to binary maps of presence/absence. For vector data, this was done by a process called rasterisation, whereby any grid cell in contact with the vector data assumes the value of the vector data. In keeping with the precautionary nature of the screening layer, we chose to rasterise based on any intersection, not requiring polygon data to cross the midpoint of the grid cell. Point data transferred their values to the grid cells they occupy. For raster data, where necessary (i.e. the data were not already binary), a classification threshold was set. Again, in keeping with the precautionary nature of the data, we set this at 0.5. This meant that any cell with >=50% of the feature was classified as a presence.

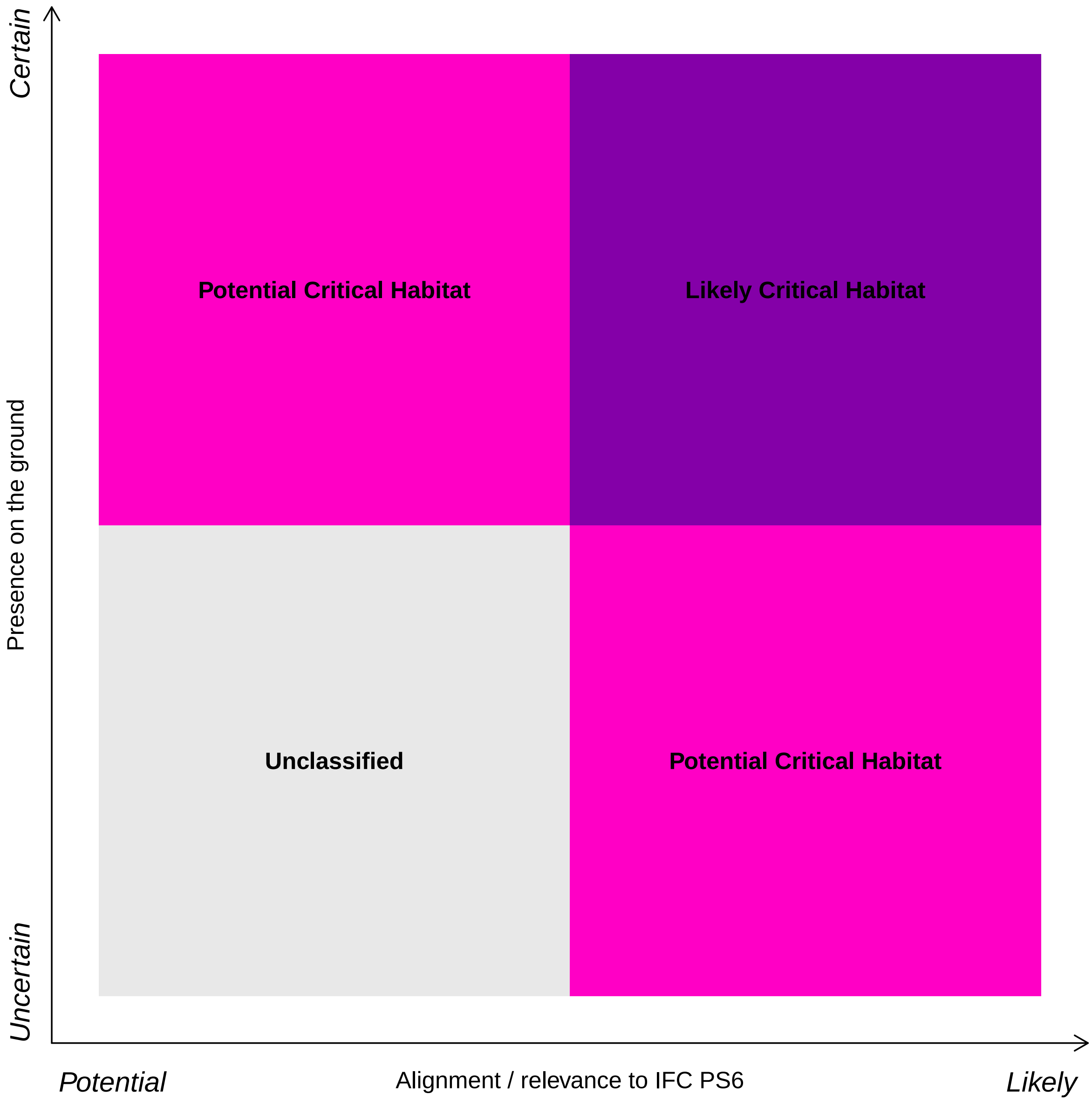

Figure 1. Classification of data as Likely or Potential Critical Habitat is based on the preceived strength of alignment with PS6 criteria and the spatial resolution of the data. Adapted from[16].

Binary raster data were then combined so that each grid cell's unique combination of biodiversity features has a unique value. Based on what features comprised the value, the grid cell was categorised hierarchically: Likely > Potential > Unclassified. See Supplementary Information for a map of unique feature combinations.

Data Records

We find that Critical Habitat, as defined by the IFC, covers 68.26 million km^2 globally (Figure 2): 51.64 million km^2 (10.2%) is Likely Critical Habitat and 16.62 million km^2 (3.2%) is Potential Critical Habitat.

These data are presented in two formats at 0.008333 degree resolution in WGS 84 projection, and represent the best publicly available global screening layer for Critical Habitat:

1. Basic Critical Habitat layer *Basic_Critical_Habitat_WGS.tif*
2. Drill down Critical Habitat layer

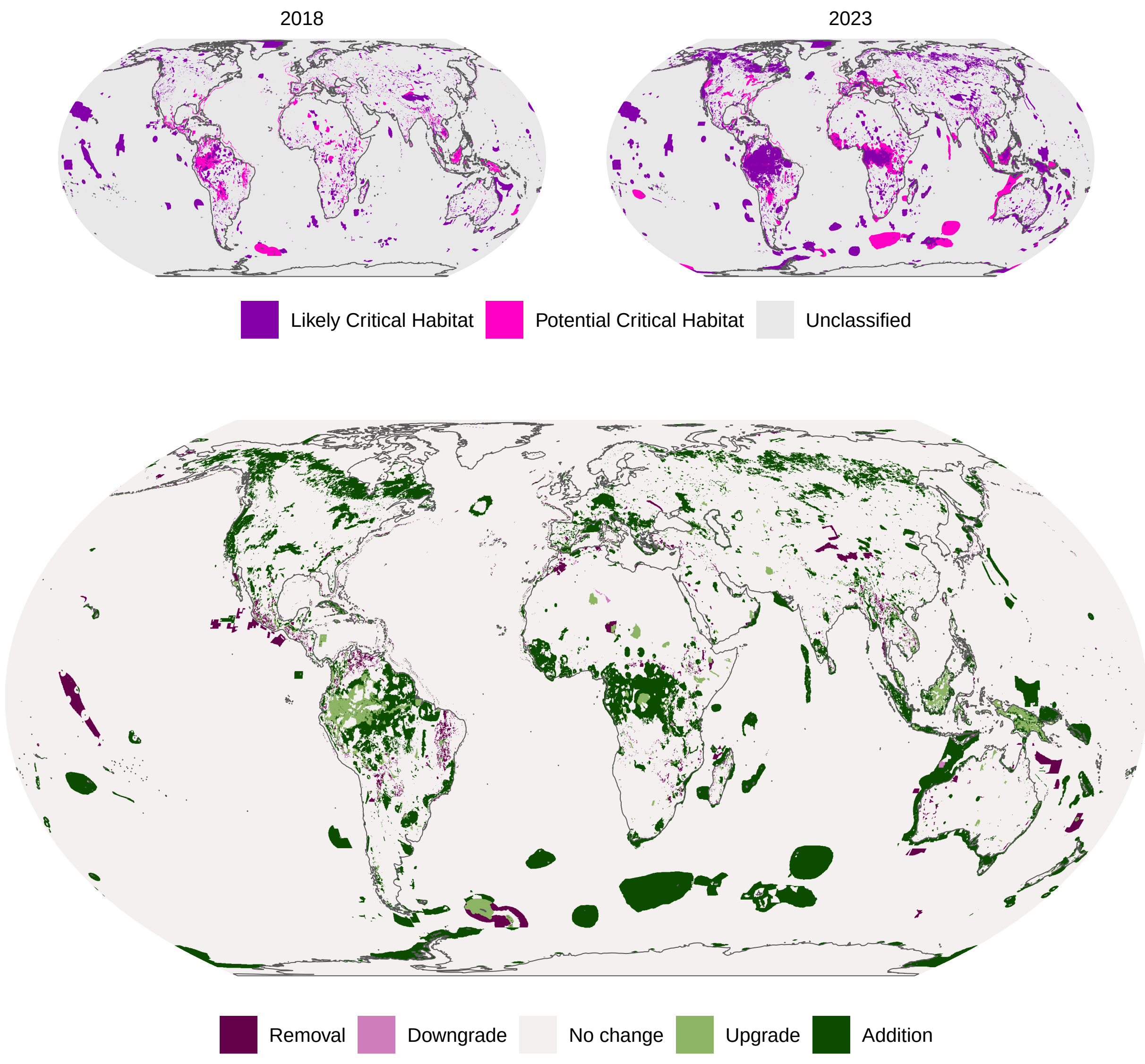

Figure 2. Global screening layer for Critical Habitat. Reprojected to Equal Earth and aggregated to 10 x 10 km for visualisation.

Technical validation

We identified 21 biodiversity feature datasets that are split into 53 separate triggers (**?@tbl-features**).

Table 2. Biodiversity features included in the analysis, their alignment with IFC PS6 Critical Habitat criteria and classification as ‘Likely’ or ‘Potential’ Critical Habitat.

		IFC PS6 criteria					
Biodiversity features	Designation criterion / Trigger	1	2	3	4	5	Classification
Cold seeps			P		L	L	Likely/Potential
Cold water corals	Modelled occurrence				P	P	Potential
	Observed occurrence				L	L	Likely
Ever-wet tropical forest					P		Potential
Hydrothermal vents	Active		L			L	Likely
	Inactive				P		Potential
Key Biodiversity Areas	Alliance for Zero Extinction Sites	L	L	L			Likely
	A1a and A1e	L					Likely
	A1b	P					Potential
	A2a				L		Likely
	A2b				P		Potential
	B1		L				Likely
	B4				P		Potential
	C				P		Potential
	D1a			L			Likely
	D1b			P			Potential
	D2			L			Likely
	D3			P			Potential
	E	P					Potential
Important Bird and Biodiversity Areas	A1	L					Likely
	A2		P				Potential
	A3				P		Potential
	A4			L			Likely
	B1b	P					Potential
Important Marine Mammal Areas	A	P					Potential
	B1		P				Potential
	B2			L			Likely
	C1, C2, and C3			P			Potential
	D1		P		P		Potential
	D2				P		Potential
Intact Forest Landscapes					L		Likely
Irreplaceable protected areas					L		Likely

(continued)

Biodiversity features	Designation criterion / Trigger	IFC PS6 criteria 1	2	3	4	5	Classification
IUCN Red List of Threatened Species	CR under criterion D	L					Likely
	EN under criterion D2	P					Potential
	Great Apes species ranges	P					Potential
Mangroves					L		Likely
Saltmarsh					L		Likely
Sea turtle nesting sites	All sea turtle species			P	P		Potential
	CR and EN sea turtle species	P					Potential
Seagrass beds					L		Likely
Seamounts					P		Potential
Tiger Conservation Landscapes		L					Likely
Tropical dry forest					P		Potential
Tropical moist forest					L		Likely
Tropical montane cloud forests					P		Potential
Warm water coral reefs					L	L	Likely
World Database on Protected Areas	All Ramsar sites				L		Likely
	Ramsar sites under criterion 2	L					Likely
	Ramsar sites under criteria 5 and 6			L			Likely
	Ramsar sites under criteria 1 and 3				L		Likely
	Ramsar sites under criteria 4, 7, 8 and 9			P			Potential
	Protected areas under IUCN management categories I and II				L		Likely
	Natural and mixed World Heritage Sites				L		Likely

Note:
’CR’: Critically Endangered; ’EN’: Endangered; ’L’: Likely; ’P’: Potential.

Table 3. Areas and percentage coverage of each Critical Habitat category in 2018 and 2023 across Areas Beyond National Jurisdiction (ABNJ), Economic Exclusion Zones (EEZ) and Land.

	2018						2023					
Domain	Likely		Potential		Unclassified		Likely		Potential		Unclassified	
ABNJ	2.4	9.2%	0.052	0.6%	220	46.4%	5	9.8%	3.2	19.2%	214	48.4%
EEZ	8.9	34.0%	2.474	24.6%	129	27.2%	14	27.0%	5.7	34.0%	121	27.4%
Land	14.9	56.8%	7.522	74.8%	125	26.4%	33	63.2%	7.8	46.8%	107	24.2%

Note:
Areas in million km^2.

Table 4. Areas and percentage cover of each IFC PS6 criteria to Potential and Likely Critical Habitat. Percentages add up to >100 due to overlapping areas.

Criteria	Likely		Potential	
1. Critically Endangered or Endangered species	26.06	50.4%	10.3	61.8%
2. Endemic and/or restricted-range species	13.20	25.6%	1.7	10.4%
3. Globally significant concentration of migratory or congregatory species	20.40	39.4%	7.0	42.2%
4. Highly threatened and/or unique ecosystems	38.15	73.8%	5.2	31.4%
5. Key evolutionary processes	0.56	1.0%	1.1	6.8%

Usage notes

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